## GETTING STARTED

Design a robot. Consider a scenario where a robot would be useful in your daily life. Identify the tasks you would like the robot to perform. Describe how the robot will work, for example, the inputs that would be used (and the hardware, e.g. sensors that might do this), how the robot will decide what to do, and what its output will be.

## WHAT IS INTELLIGENCE?

There are many ideas on what it is to be intelligent, with one key area being the ability to learn. How did you develop intelligence? From when you were very young you started to learn. Each time you experienced something new, you learnt from it, for example, if you cried then someone might pick you up. So, next time you wanted to be picked up, you cried.

The desire to create intelligent computers has been around since the very first machines. Even looking back to ancient history there are stories about bringing objects to life.

Achieving artificial intelligence has made the most steps over the last 20 to 30 years. There are now systems that simulate some human activities, and systems that can learn both with and without a person telling them what to learn. These programs all still run on rules, that are programmed into them to follow, so are they intelligent? Humans learn by developing rules and storing these in memory, so are we intelligent?

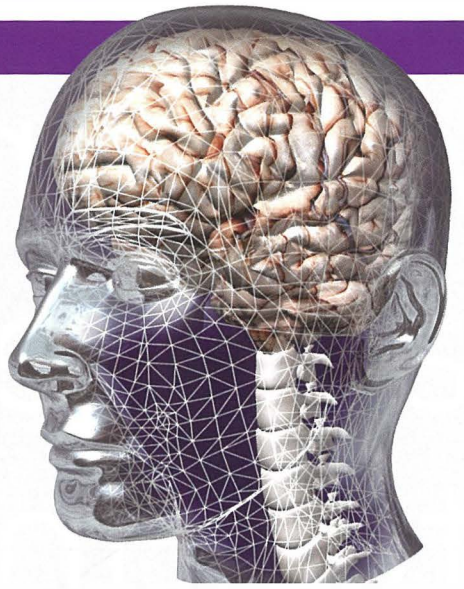


**Figure 6.1:** A brain

### Discussion questions

1 What do you think is meant by intelligence?

2 How would you test whether a computer is intelligent, or not? What would you test it against?

3 Do you think that computers that learn and act independently should be created?

# 6.1 Automated systems

An **automated** system performs actions without interaction with humans. They are used in small systems such as the heating or air-conditioning in buildings, all the way up to flying aeroplanes and building items such as cars.

**KEY WORD**

**automated:** a device that is operated without human interaction.

## Features of automated systems

An automated system usually has the following features:

- Sensors – to measure the environment.
- Microprocessor – to process the data from the sensors and determine if an action needs to be taken.
- Actuators – to create a movement.

There are a lot of different **sensors** that you need to know about – you can find more information on these where they were covered in Chapter 3, Section 6. You will need to identify which sensor is most appropriate in a given context.

For example: a system needs to monitor the pollution in a river.

Consider what could be measured. The temperature of the water using a temperature sensor. The amount of light reaching the bottom using a light sensor. The pH of the water using a pH sensor.

Sensors just do one job, they read the environment and send the data onwards. They do not process the data. That is done by the **microprocessor**. There is a problem though, the sensor reads **analogue** data (real-world data such as 23.5 °C), but the microprocessor only understands binary (01101010). The sensor first sends the data to an analogue-to-digital converter, to turn the reading into binary (digital). Then, the **digital** data is sent to the microprocessor.

Before the system is started, data will be stored in the microprocessor. This could be a single value, for example, the value from an infra-red sensor to state that there is movement. There could be several values, for example, the minimum temperature and maximum temperature for a greenhouse. Or there could be a whole range of data.

The microprocessor takes the data from the sensor and compares it to its value, or values. The result of the comparison determines the next step. For example, in a burglar alarm, if the sensor is outside of the stored range then an alarm will need turning on. In a house, if the temperature is below the minimum the heating turns on, and if it's above the maximum it turns the heating off.

If the microprocessor decides an action is needed (e.g. turn the heating on), it sends a signal to the **actuator**, which then performs the action. The whole process is then repeated until the system is turned off.

KEY WORDS

**sensor:** a type of input device that is used to capture data from its immediate environment.

**microprocessor:** an integrated circuit that is able to perform many of the functions of a CPU.

**analogue:** continuous data that can be any value within a range.

**digital:** discrete data that is stored as 1s and 0s.

**actuator:** a mechanical part that causes another device or part to move.

## Example

A set of doors in a building automatically open when a person is detected.

1 What type of sensor could be used?

An infra-red to detect movement, or a pressure sensor to detect if someone is standing near it.

2 What happens to the analogue data?

The analogue data is converted into digital.

3 Where does the data go?

The data is sent to the microprocessor.

4 What happens in the microprocessor?

It compares the received data with the stored data. If it is out of range, it sends a signal to an actuator to open the doors. If it is not out of range, it does nothing.

This process continues until the system is turned off.

This can also be shown in a diagram in Figure 6.2:

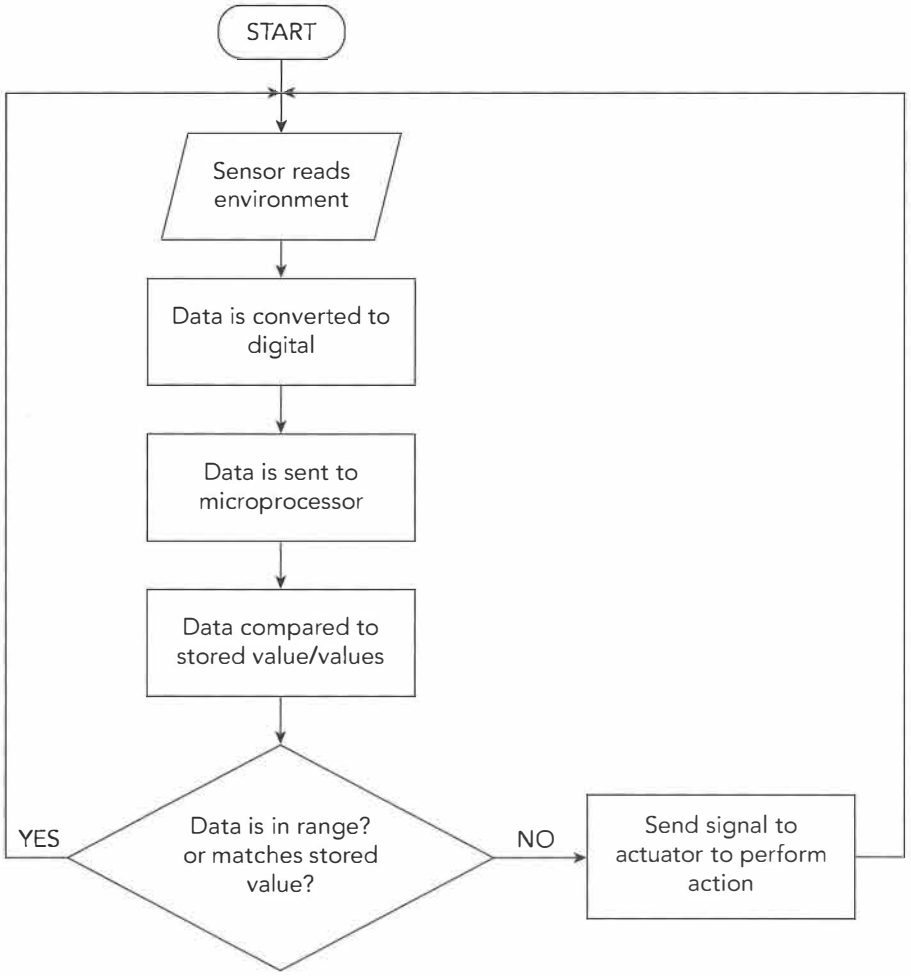



**Figure 6.2:** The process in which doors open automatically when a person is detected

## COMPUTER SCIENCE IN CONTEXT

Control systems are at work all around you, probably without you noticing. Think about a car. It could have light sensors, so when it is getting dark the headlights will be automatically turned on. It could have rain sensors, so when it starts raining the windscreen wipers start moving. These each have a sensor that is constantly reading the values of the environment, e.g. the light level. The data will be sent to a microprocessor that compares the value with its stored value, and if necessary sends the signal to turn on the lights, or the wipers. This means the car drivers do not have to worry about turning these on when needed. It also improves safety, particularly with the headlights so that the car is always visible.

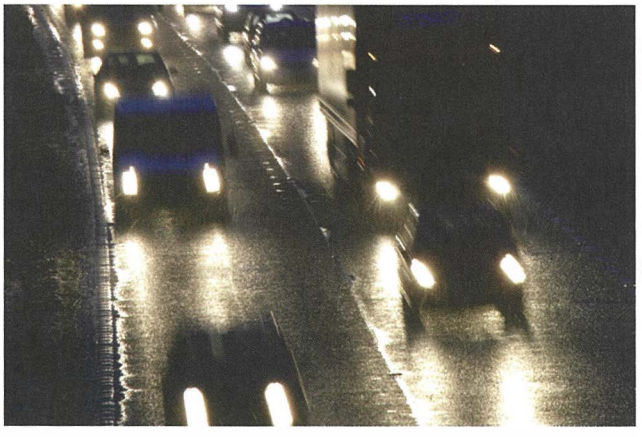


**Figure 6.3:** Car headlights could operate via sensors

SKILLS FOCUS 6.1

## CONTROL AND MONITORING SYSTEMS

When considering a control or monitoring system you need to determine the inputs, processes and outputs as you would in a computer program:

- The inputs from the sensors.
- The processes.
- The outputs from the system.




**Figure 6.4:** A control or monitoring system can be thought of as a 'story' of input, process and output

Using these you then need to create the 'story' (see Figure 6.4) about how the sensors gather the appropriate data, what this is compared to, and how this influences the outputs.

For example:

An automatic door will open when a person is detected. The door stays open for 5 seconds if no more people are detected.

Take each point in turn:

- Inputs:

  A pressure sensor can be used to detect if a person is standing on a specific point.
  A motion or infra-red sensor can be used to detect if a person is in a specific place.

- Processes:

  Analogue data from the sensor will be converted to digital.
  The microprocessor will store the value where a person is detected on the pressure sensor, and/or the result identifying that a person (movement) has been detected.

- Output:

  A signal will go to an actuator to open the door.
  This will stay for 5 seconds, and is reset each time another person is detected.

### Questions

1 Describe how the sensor will monitor its environment.
2 Describe how the processor will work out if the door needs to be opened.
3 When will the door close?

## Questions

1 What type of data does a microprocessor process?
2 What tasks does a sensor perform?
3 An automated system makes cakes by positioning a cake tin below a funnel where the cake mix comes through to fill the tin.
   a Identify a suitable sensor for this system.
   b The sensor has read the data. Identify the function that happens next.
   c Where is the data sent?
   d Describe the actions the microprocessor performs.
   e How many times does the system run?

## Automation in context

KEY WORD

**automation:** the use of automated equipment in a process, e.g. a factory.

The **automation** of systems has both benefits and drawbacks, and these are different depending on the system. You need to be able to think about the context and the benefits and drawbacks in that scenario.

When evaluating the use of automation you can consider:

- initial cost
- running cost
- safety
- replacing people's jobs
- continuous work all day every day
- precision.

### Example 1, industry

Car manufacturing.

- Initial cost: this will be high because the robots need purchasing. The software will also need developing and could be expensive.
- Running cost: this could be high because the systems need maintenance, but it could perform actions more efficiently than humans which could save money. However, there may be fewer employees, therefore saving money long-term.
- Safety: it should be safer because they can monitor their environment and stop if there is any danger, whereas people can be distracted, and more accidents could happen.



**Figure 6.5:** A car manufacturing factory

- Replacing people's jobs: jobs that the system now performs will be lost, but more jobs will be made to maintain the system.
- Continuous work all day every day: this would mean that more cars can be built.
- Precision: human errors will not be made so there are less faulty cars.

## Example 2, science

Nuclear power plant.

- Initial cost: will be high because of the purchase of the devices and development of the system.
- Running cost: may be lower because fewer people are needed, however maintenance might be expensive.
- Safety: less risk of human error therefore increasing the safety, also people will not need to work in dangerous places.
- Replacing people's jobs: some jobs will be replaced for the functions the system does, but new jobs will be made to maintain the system.
- Continuous work all day every day: it is important that the power plant is monitored all day every day without getting tired and therefore less errors.
- Precision: this is important in a high-risk scenario and an automated system will measure and perform actions with more precision, therefore fewer errors.

## Example 3, agriculture

Growing conditions within a greenhouse.

- Initial cost: this may be high for the installation and programming.
- Running cost: there will be a cost for electricity and maintenance but it is a small system so may not be high.
- Safety: this may not be relevant because it is not a dangerous process.
- Replacing people's jobs: it might replace jobs of people who monitored the conditions, but this is unlikely to be a large number.
- Continuous work all day every day: this will allow the system to adapt as soon as there is a problem, instead of waiting for a person to check and make changes.
- Precision: this may not be as important, but it will allow readings to be more accurate and instant changes to be made.



**Figure 6.6:** A greenhouse

# Questions

4 An auto-pilot system will fly an aeroplane without input from a person.
Identify **two** benefits and **two** drawbacks of using the auto-pilot.

5 A weather station records and stores data about the current weather conditions.
Describe the advantages and disadvantages of using the automated system.